Recovering Mass–Energy Equivalence within Algebraic Lorentzian Read-out

A Conditional Theorem in Q-RAIF A

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Abstract

This paper establishes a conditional recovery theorem for mass–energy equivalence within the Q-RAIF A programme. Revised Q-RAIF A is used as modelling motivation for a separately supplied effective Lorentzian read-out with signature $(-,+,+,+)$; the present paper does not derive that read-out from algebraic constraints. The bridge input is formulated on the effective read-out representation rather than on the global source algebra: an admissible local translation action is a strongly continuous unitary representation $U(a)$ of local spacetime translations on the read-out sector, and admissible four-momentum is the self-adjoint Stone generator $\hat{P}_\mu$ of this representation, with $U(a) = \exp(-ia^\mu \hat{P}_\mu/\hbar)$. A c-number four-momentum P^μ is then obtained as a spectral, semiclassical, or sharply supported read-out of these generators. Under this bridge assumption and its consistency requirements, the standard Lorentzian mass–energy relation follows as a theorem: $m^2c^2 = -g_{\mu\nu}P^\mu P^\nu$, $E_u = -c\, g_{\mu\nu}P^\mu u^\nu$ for a dimensionless unit observer direction u, $E_u^2 = m^2c^4 + |\mathbf{p}_u|^2c^2$, and, in the rest frame, $E = mc^2$.

The theorem-level claim is intentionally narrow. It is a recovery result, not a derivation of special relativity from nothing and not a derivation of Standard Model masses, the Higgs mechanism, QCD confinement, or the numerical value of c. Its contribution is structural: once a Lorentzian read-out and an admissible momentum representation have been supplied, mass–energy equivalence need not be reinstated as an independent postulate but is recovered inside that read-out. Programme-level readings of c^2, Higgs/QCD mass contributions, and vacuum sectors are isolated in a separate part of the paper and do not extend the theorem-level claim.

Keywords: Q-RAIF A, Lorentzian emergence, mass–energy relation, four-momentum, modular flow, admissible read-out, Higgs mechanism, QCD, structural imprint

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1 Introduction

The formula $E = mc^2$ is often introduced as a slogan about conversion: mass can become energy, and energy can become mass. That slogan is useful pedagogically, but it hides the more precise geometric statement. In relativistic mechanics, mass and energy are not two independent substances tied together by an arbitrary conversion factor. They are related readings of a single four-momentum object in a Lorentzian geometry. The invariant mass is the Lorentzian norm of four-momentum; the energy is the component of that same four-momentum relative to an observer's time direction. The constant c is not merely a numerical convenience. It is the scale relating temporal and spatial units in the Lorentzian metric, and the same scale fixes the light cone.

Within the Q-RAIF A programme, this observation has a natural but delicate status. Revised Q-RAIF A supplies a Clifford representation as a model for a separately specified Lorentzian read-out; it does not derive Lorentzian geometry from its algebraic constraints [20]. Once a Lorentzian read-out is assumed, the standard mass–energy relation follows by ordinary relativistic geometry, provided an admissible four-momentum is also defined. The present paper formalizes exactly that conditional step.

The claim is a conditional convention check. Given a Lorentzian read-out, a four-vector P^μ, and the definition $m^2c^2 = -g(P,P)$, the mass-shell identities follow algebraically. The physical content lies entirely in the two bridge inputs: existence of the Lorentzian effective sector and existence of a well-posed translation/Stone representation. Neither is derived here from Tomita–Takesaki data, and no Standard Model mass spectrum is derived.

The distinction matters. Modular flow does not force four-momentum, particle spectra, Higgs/Yukawa couplings, or QCD binding energies. The Lorentzian arena and momentum object are both supplied as bridge data; the paper then checks the usual mass–energy identity.

1.1 The epistemic split

The paper separates three levels.

Level	Content	Status
A	The effective sector is supplied with a Lorentzian read-out modelled in Q-RAIF A language	Interface assumption, not a Q-RAIF A theorem
B	Admissible four-momentum is the Stone-generator read-out of a strongly continuous admissible local translation representation on the effective sector	Bridge assumption with consistency checks
C	$m^2c^2 = -g(P,P)$, $E_u = -c\,g(P,u)$, $E_u^2 = m^2c^4 + \lvert\mathbf{p}_u\rvert^2c^2$, and rest-frame $E = mc^2$	Definitional Lorentzian identity

Level C is a standard identity once Levels A and B and the invariant-mass definition are supplied. Level A is an interface assumption summarized in Section 2. Level B is the load-bearing bridge assumption. Its consistency checks are verified below only in the flat standard limit; their joint realization in an emergent Type III_1 read-out sector remains open.

1.2 What the paper does not claim

The scope is deliberately narrow.

- It does not derive special relativity from nothing.
- It does not replace the standard Lorentzian derivation of the mass–energy relation.
- It does not derive the existence of the Standard Model, the Higgs mechanism, QCD confinement, or numerical particle masses.
- It does not claim that c^2 is literally identical with a substance, object, or hidden field.
- It does not allow Ω to split into multiple actual states or to be replaced by Ω'.
- It does not identify Higgs/Yukawa mass and QCD binding energy as the same physical mechanism.

The contribution is architectural: it displays exactly which bridge assumptions are sufficient to place the standard mass–energy identity inside the programme. It neither alters special relativity nor converts the revised Q-RAIF A model choice into an emergence theorem.

2 Background: Q-RAIF A Lorentzian Read-out

The present paper makes the following interface assumption, using Q-RAIF A only as modelling motivation.

Assumption 2.1 (Effective Lorentzian read-out interface)**.** A finite effective sector selected from the source pair $(\mathcal{A}_U, \Omega)$ is supplied with a local Lorentzian description of signature $(-,+,+,+)$ on the domain where the read-out approximation is valid.

Revised Q-RAIF A gives a possible Clifford encoding of this supplied geometry but does not prove the assumption. Its role here is solely to state the arena in which the mass–energy identities are checked.

Remark 2.2 (No hidden reproof)*.* Nothing in the proof of Theorem 4.3 below establishes Assumption 2.1. Conversely, that assumption alone does not supply a four-momentum object. This paper therefore introduces admissible four-momentum as a separate bridge definition in Section 3.

2.1 Self-contained summary of the Q-RAIF A input

For external readability, we record the modelling inputs used here. This is not a summary of a Q-RAIF A proof; it is the dependency chain assumed by the present convention check.

(Q1) **Source data.** The starting data are a source algebraic domain $\mathcal{A}_U$ and a single faithful normal state Ω. In the broader programme, the pair $(\mathcal{A}_U, \Omega)$ supplies the algebraic environment in which finite read-out sectors are selected.

(Q2) **Modular asymmetry.** Under standard Tomita–Takesaki hypotheses, the state Ω determines a modular automorphism group σ_t^{Ω}. Q-RAIF A uses this as the algebraic marker of a distinguished direction in the read-out construction. The present paper does not use modular flow to define momentum.

(Q3) **Admissibility.** A finite read-out is admissible only if it preserves compositional consistency, metric compatibility, and causal-channel distinguishability in the effective sector. Q-RAIF A formulates these as algebraic constraints on the read-out rather than as background spacetime assumptions.

(Q4) **Signature input.** The effective read-out is required to carry an indefinite metric; with observed local dimensionality, the chosen signature is $(-,+,+,+)$. This is not derived from the positive information metric.

(Q5) **Effective geometry.** The output used here is a local effective Lorentzian read-out (M_{eff}, g) on the accessible sector. This output supplies the metric structure for the theorem below, but it does not by itself supply particle spectra, translation generators, or Standard Model mass mechanisms.

The present paper uses item (Q5) as an interface input and introduces momentum separately in Section 3. The Lorentzian read-out assumption and the momentum bridge are logically distinct.

2.2 Single global state

The notation $(\mathcal{A}_U, \Omega)$ is used throughout in the same sense as the broader programme. $\mathcal{A}_U$ denotes the source algebraic domain and Ω denotes a single faithful normal state used to define the relevant algebraic read-out structure. The present paper never introduces multiple actual Ω's. In particular, changes of effective vacuum sector, phase, or low-energy spectrum are not written as $\Omega \to \Omega'$.

Assumption 2.3 (Effective read-out sector)**.** Let $\mathcal{R}_{\text{adm}}$ be an admissible finite read-out sector of $(\mathcal{A}_U, \Omega)$ satisfying the hypotheses of Q-RAIF A. The sector carries an effective Lorentzian manifold description (M_{eff}, g) with signature $(-,+,+,+)$ on the domain where the finite read-out approximation is valid.

The phrase "effective Lorentzian manifold description" is not meant to erase the algebraic origin of the geometry. It means only that, at the read-out scale under discussion, the accessible variables admit a Lorentzian geometric representation adequate for the usual local tensorial constructions.

2.3 Locality and the finite-read-out regime

The mass–energy theorem below is local. It is stated in a domain where a local orthonormal frame can be chosen and where effective translation generators make sense within the admissible read-out. This is the same modesty present in ordinary relativistic physics: four-momentum is local or asymptotic unless additional global symmetries are supplied.

For curved effective geometries, the theorem should be read in a local inertial frame or in a region admitting the relevant approximate translation symmetry. The theorem does not require global Minkowski spacetime.

3 Admissible Four-Momentum

The step from effective Lorentzian geometry to a mass–energy relation requires an object that can be called four-momentum. Q-RAIF A motivates the Lorentzian read-out, but in the revised

programme that arena is supplied as an interface assumption; it does not, by itself, force a particle momentum structure. The present paper therefore introduces the bridge explicitly.

Definition 3.1 (Admissible local translation action)**.** Within an admissible effective read-out sector $\mathcal{R}_{\text{adm}}$, an *admissible local translation action* is a local action

$$\alpha_a : \mathcal{R}_{\text{adm}} \to \mathcal{R}_{\text{adm}}, \qquad a \in \mathcal{O} \subset \mathbb{R}^{1,3},$$

where $\mathcal{O}$ is a sufficiently small neighbourhood of the origin in the effective tangent chart, such that:

(i) α_a acts within the finite read-out domain to the order of approximation used;
(ii) α_a preserves the effective Lorentzian structure and admissibility conditions to that order;
(iii) $\alpha_{a+b} = \alpha_a \circ \alpha_b$ whenever $a, b, a+b \in \mathcal{O}$;
(iv) in a local inertial or large-system limit, α_a reduces to the usual local spacetime translation action of relativistic mechanics.

Assumption 3.2 (Unitary implementation on the read-out representation)**.** The admissible local translation action is assumed to be implemented on the effective read-out representation Hilbert space $\mathcal{H}_{\mathcal{R}_{\text{adm}}}$ by a strongly continuous local unitary representation

$$U(a) : \mathcal{H}_{\mathcal{R}_{\text{adm}}} \to \mathcal{H}_{\mathcal{R}_{\text{adm}}}, \qquad \alpha_a(A) = U(a) A U(a)^*$$

for read-out observables A in the represented sector. The representation is local or approximate in the same sense as the effective Lorentzian chart; no global translation symmetry of $\mathcal{A}_U$ is assumed.

Definition 3.3 (Admissible four-momentum)**.** Given Assumption 3.2, the *admissible four-momentum operators* $\hat{P}_\mu$ are the self-adjoint Stone generators of the strongly continuous local translation unitaries:

$$U(a) = \exp\left(-\frac{i}{\hbar} a^\mu \hat{P}_\mu\right)$$

on the common invariant domain where the expression is defined. A c-number four-momentum P^μ used in the mass–energy theorem is a spectral, semiclassical, expectation-value, or sharply supported read-out of these generators in a local Lorentzian frame, transforming as a Lorentz vector under admissible local frame changes.

This definition is intentionally conservative. It does not claim that $\hat{P}_\mu$ has been derived directly from modular flow or that global translation generators exist on $\mathcal{A}_U$. It states the momentum object required to formulate relativistic dynamics inside an effective Q-RAIF A Lorentzian read-out once a local translation representation is available.

3.1 Consistency requirements

Definition 3.3 is acceptable only if four requirements are met.

Assumption 3.4 (Well-posedness of admissible four-momentum)**.** The admissible four-momentum definition is required to satisfy the following conditions. These requirements are non-trivial because the effective read-out sector is finite, local, or approximate; they are not automatically guaranteed by the abstract Stone theorem.

(a) **Existence**: an admissible local translation action and its strongly continuous unitary implementation exist in the read-out sector or in the local/asymptotic regime under discussion;

(b) **Stone-generator well-posedness**: the self-adjoint generators $\hat{P}_\mu$ exist on a common invariant domain adequate for the local read-out calculation;

(c) **Covariance up to admissible frame transformation**: the generator read-outs transform as four-vector components under the Lorentz transformations allowed by the effective geometry;

(d) **Standard-limit recovery**: when the admissibility restriction is relaxed to the standard local relativistic regime, P^μ agrees with the usual special-relativistic four-momentum.

These are not cosmetic requirements. Without them, the symbol P^μ would be an unsupported import from ordinary physics. With them, it becomes a controlled bridge object: defined in the Q-RAIF A read-out, constrained by the effective Lorentzian geometry, and checked against the standard relativistic limit.

The four conditions are verified in this paper only in the standard flat-Minkowski limit of Appendix B. Their satisfiability in an emergent Type III_1 read-out sector is an open problem. In particular, admissibility has not been shown to produce a sector supporting strongly continuous local translation unitaries with self-adjoint generators on a common invariant domain. These existence, Stone, covariance, and standard-limit conditions belong to this Mass–Energy companion; they are not a result of Q-RAIF A or a nonexistent Q-RAIF A Appendix B.

Remark 3.5 (Why this is a definition rather than a lemma). One could try to derive admissible translation unitaries and their Stone generators from deeper algebraic data. That result is not available. The present paper introduces the local translation representation as a bridge assumption and records the standard identity that follows once it is admitted.

Remark 3.6 (Why the construction is not made on the global Type III source algebra). The global source algebra $\mathcal{A}_U$ may be Type III and need not carry a trace, a global position operator, or a global translation representation. The present construction therefore does not define momentum as a global conjugate variable inside $\mathcal{A}_U$. It defines momentum only after an effective read-out representation supporting local/asymptotic translation unitaries has been assumed. Whether admissibility actually produces such a representation is not established; this is the main open technical bridge.

3.2 Relation to modular flow

Tomita–Takesaki theory supplies, under standard hypotheses, a modular automorphism flow associated with a faithful normal state. The present paper separately assumes an effective Lorentzian geometry through Assumption 2.1. It does not claim that modular flow determines the full four-momentum object or the particle content of the effective theory.

4 The Conditional Mass–Energy Theorem

This section states and proves the theorem-level content of the paper.

4.1 Conventions

Let (M_{eff}, g) be an effective Lorentzian read-out with signature $(-,+,+,+)$. In local orthonormal coordinates we write

$$g_{\mu\nu} = \text{diag}(-1, 1, 1, 1), \tag{1}$$

when using $x^0 = ct$, and equivalently

$$ds^2 = -c^2 dt^2 + d\mathbf{x}^2 \tag{2}$$

when using the coordinate time t. The four-momentum is written

$$P^\mu = \left(\frac{E}{c}, \mathbf{p}\right) \tag{3}$$

in a local inertial frame.

For an observer, we use a dimensionless future-directed unit timelike vector u^μ satisfying

$$g_{\mu\nu} u^\mu u^\nu = -1. \tag{4}$$

The energy measured by that observer is

$$E_u \equiv -c\, g_{\mu\nu} P^\mu u^\nu. \tag{5}$$

With this convention, an observer at rest in the local frame has $u^\mu = (1, 0, 0, 0)$ and $E_u = E$.

Definition 4.1 (Invariant mass)**.** For an admissible four-momentum P^μ in the effective Lorentzian read-out, define the invariant mass m by

$$m^2 c^2 \equiv -g_{\mu\nu} P^\mu P^\nu. \tag{6}$$

Definition 4.2 (Observer energy and spatial momentum)**.** For a dimensionless unit observer direction u^μ, define

$$E_u \equiv -c\, g(P, u). \tag{7}$$

The observer-relative spatial momentum $\mathbf{p}_u$ is the projection of P^μ onto the spacelike hypersurface orthogonal to u^μ.

Theorem 4.3 (Conditional mass–energy relation)**.** *Assume the effective Lorentzian read-out interface (Assumption 2.1), an effective sector $\mathcal{R}_{\text{adm}}$, a local translation representation satisfying Assumption 3.2, and a c-number four-momentum read-out P^μ obtained from Definition 3.3. Then, in any local orthonormal frame of the effective Lorentzian geometry,*

$$E^2 = m^2 c^4 + |\mathbf{p}|^2 c^2. \tag{8}$$

Equivalently, relative to an observer direction u,

$$E_u^2 = m^2 c^4 + |\mathbf{p}_u|^2 c^2. \tag{9}$$

In the rest frame, where $\mathbf{p} = 0$, this reduces to

$$E = mc^2. \tag{10}$$

Proof. In a local orthonormal frame with signature $(-,+,+,+)$, write

$$P^\mu = \left(\frac{E}{c}, p^1, p^2, p^3\right).$$

Then

$$g_{\mu\nu}P^\mu P^\nu = -\left(\frac{E}{c}\right)^2 + |\mathbf{p}|^2.$$

By Definition 4.1,

$$m^2c^2 = -g_{\mu\nu}P^\mu P^\nu = \left(\frac{E}{c}\right)^2 - |\mathbf{p}|^2.$$

Multiplying by c^2 gives

$$m^2c^4 = E^2 - |\mathbf{p}|^2c^2,$$

hence

$$E^2 = m^2c^4 + |\mathbf{p}|^2c^2.$$

The observer-relative form follows by choosing a local orthonormal frame adapted to u, in which $u = (1,0,0,0)$ and the spatial components of P are precisely $\mathbf{p}_u$. If $\mathbf{p} = 0$ in the rest frame, the relation reduces to $E^2 = m^2c^4$. For positive-energy future-directed momentum, $E = mc^2$. □

Corollary 4.4 (Massless read-out)**.** *If $m = 0$, then P^μ is null and*

$$E = |\mathbf{p}|c. \tag{11}$$

Proof. Set $m = 0$ in Theorem 4.3. Then $E^2 = |\mathbf{p}|^2c^2$, and the positive-energy branch gives $E = |\mathbf{p}|c$. Equivalently, $g(P,P) = 0$, so P^μ lies on the null cone. □

Remark 4.5 (What is theorem-level here). The calculation is standard and, given Level B, Level C is definitional: invariant mass is defined by the Lorentzian norm of the supplied four-momentum. The result records that the bridge is sufficient and fixes conventions. Its physical content lies entirely in the unproved read-out and translation/Stone bridge assumptions.

5 Programme-Level Readings

Sections 5.1–5.3 are programme-level readings of the theorem-level result. They do not enter the proof of Theorem 4.3, do not extend its mathematical claim, and do not replace the standard physics of special relativity, quantum field theory, the Higgs mechanism, QCD, or vacuum decay. Their purpose is to locate the recovered mass–energy relation within the broader Q-RAIF A and imprint vocabulary.

5.1 The Role of c^2

Theorem 4.3 already contains the mathematical role of c. This section clarifies what can be said at theorem level and what belongs to programme-level interpretation.

5.1.1 Theorem-adjacent facts

In the local form

$$ds^2 = -c^2 dt^2 + d\mathbf{x}^2, \tag{12}$$

the constant c sets the relative scale between coordinate time and spatial distance. The null condition $ds^2 = 0$ gives

$$|\mathrm{d}\mathbf{x}| = c\,|\mathrm{d}t|, \tag{13}$$

so the same constant fixes the light-cone slope. In the rest-frame mass–energy relation,

$$E = mc^2, \tag{14}$$

the same scale appears squared because energy and mass are related through the squared norm of four-momentum.

Thus three familiar appearances of c are mathematically connected:

Appearance	**Lorentzian role**
Metric coefficient	$ds^2 = -c^2 dt^2 + d\mathbf{x}^2$ fixes temporal-spatial scaling
Null cone	$ds^2 = 0$ gives $\lvert\mathrm{d}\mathbf{x}\rvert/\lvert\mathrm{d}t\rvert = c$
Rest energy	$E = mc^2$ follows from the Lorentzian norm of P^μ in the rest frame

This is not a claim that c^2 is a substance or a hidden object. It is a statement about the role of a scale parameter in Lorentzian geometry.

5.1.2 Programme-level reading of c^2

The following discussion provides programme-level structural readings of Theorem 4.3's content. These readings are interpretive and do not extend the theorem-level claim.

Within the Q-RAIF A programme, the effective Lorentzian metric is treated as a read-out structure rather than a brute background. In the present paper, however, that metric is supplied as an interface assumption, not derived under admissibility constraints from $(\mathcal{A}_U, \Omega)$. On that reading, c^2 may be viewed as the scale at which the read-out distinguishes temporal from spatial directions while still placing them in a single quadratic form. It is the same scale that allows timelike, spacelike, and null directions to be compared within one geometry.

This gives the main structural angle of the paper. The appearances of c in mass–energy equivalence, null propagation, and causal-cone aperture are not treated here as three unrelated constants that happen to have the same value. Under the single effective Lorentzian metric assumed in this paper, they are three attestations of the same metric scale. This statement should not be overread as an independent phenomenological prediction: if one allows multi-metric, dispersive, Lorentz-violating, or non-standard effective structures, the theorem must be restated with those structures made explicit. Within the present single-metric Lorentzian read-out, however, the three roles of c have a common geometric source.

This motivates a disciplined structural sentence:

> Mass and energy are not two independent substances connected by an arbitrary conversion factor; they are observer-invariant and observer-relative readings, respectively, of the same admissible four-momentum object in an effective Lorentzian read-out.

More precisely, mass is the invariant norm reading of P^μ, while energy is the observer-time component reading of P^μ. The phrase "same object" refers to P^μ in the effective geometry. It does not refer to an unobserved metaphysical object behind the mathematics.

5.1.3 Corrected formulation

The formulation just given avoids a common inversion. It would be incorrect to say that mass is the timelike projection and energy is the object itself. In standard Lorentzian mechanics, the invariant mass is determined by the norm of P^μ, while energy is a frame-dependent component. The correct reading is: mass is the Lorentzian-invariant norm of admissible four-momentum, while energy is the frame-dependent timelike component relative to an observer direction. The corrected statement is:

> The admissible four-momentum P^μ is the complete Lorentzian momentum readout. Its invariant norm is called mass. Its component along an observer's timelike direction is called energy. In the rest frame, the component reading exhausts the invariant contribution and yields $E = mc^2$.

This correction strengthens the programme. It prevents the interpretive layer from overriding the physical definitions it is meant to clarify.

5.2 Physical Anchoring: Higgs/Yukawa and QCD Contributions

The theorem in Section 4 concerns the Lorentzian relation between four-momentum, invariant mass, and energy. It does not derive the physical origin of the mass values appearing in the effective theory. Those origins belong to quantum field theory and the Standard Model. Still, a physical anchor is useful because it prevents the structural reading from floating free of known physics.

5.2.1 Two different mechanisms

In the Standard Model, elementary fermion masses arise through Yukawa couplings to the Higgs field after electroweak symmetry breaking. Schematically, a fermion mass term has the form

$$m_f = \frac{y_f v}{\sqrt{2}}, \tag{15}$$

where y_f is the Yukawa coupling and v is the Higgs vacuum expectation value.

By contrast, most of the mass of ordinary baryonic matter is not the sum of elementary quark rest masses. The proton and neutron masses arise predominantly from QCD dynamics: gluon fields, quark kinetic energy, confinement, and the trace anomaly. A common pedagogical summary is that the Higgs/Yukawa contribution accounts for only a small fraction of ordinary nucleon mass, while QCD dynamics accounts for the overwhelming majority. The exact decomposition is scheme-dependent, but the qualitative distinction is robust.

Contribution	**Physical mechanism**	**Role in this paper**
Elementary rest masses	Higgs vacuum expectation value plus Yukawa couplings	Supplies elementary mass parameters in the effective theory
Nucleon mass bulk	QCD confinement, gluon energy, quark kinetic energy, trace anomaly	Supplies composite mass through strongly interacting field dynamics

These mechanisms must not be physically collapsed into one another. Higgs/Yukawa mass and QCD binding contributions are different mechanisms in Standard Model physics.

5.2.2 Programme-level read-out placement

The following reading is interpretive and does not extend Theorem 4.3. Once the physical distinction is kept intact, the two mechanisms can be placed at different depths of the same broad read-out chain. Higgs/Yukawa couplings set elementary mass parameters after electroweak symmetry breaking. QCD dynamics then generates most of the stable mass of ordinary hadronic matter through confinement-scale field dynamics. Both contributions enter the same Lorentzian mass–energy bookkeeping because both contribute to the invariant norm or internal energy content of effective stable structures.

In this limited sense, one may say:

> Higgs/Yukawa mass and QCD binding energy are different physical mechanisms, but both become legible as mass contributions inside the same effective Lorentzian read-out.

That sentence is weaker than saying they are physically the same. It is also stronger than treating the $1\%/99\%$ pedagogical split as mere trivia. It explains why different physical mechanisms can be read by the same invariant-mass bookkeeping once the effective Lorentzian sector is in place.

5.2.3 The "matter is not solid stuff" lesson

The familiar popular claim that matter is "mostly energy" should be handled carefully. It is not correct to say that matter simply does not exist. A better statement is that ordinary matter is not made of tiny solid pellets whose mass is an intrinsic lump of substance. It is a stable field-theoretic structure whose effective mass includes rest-mass terms, kinetic contributions, field energy, confinement dynamics, and stress-energy structure.

In the vocabulary of this paper, matter is an effective stable read-out carrying invariant mass within a Lorentzian sector. That description is compatible with standard physics. It does not replace it.

5.2.4 Counterfactual removal of the strong force

Statements such as "turning off the strong force would reduce a human body's mass by approximately 99%" should be treated only as informal intuition pumps. If the strong interaction were removed, nuclei and hadrons would not remain as stable objects whose masses could simply be reweighed. The relevant structure would collapse, not merely become lighter. The correct lesson is that the mass of ordinary matter depends deeply on QCD stabilization; it is not that one can subtract a stable component while leaving the rest of the material hierarchy intact.

5.3 Vacuum Sectors Without Splitting Ω

The Higgs mechanism and discussions of vacuum metastability introduce another possible confusion. A change in effective vacuum sector must not be written as a change from Ω to Ω' in the present framework.

Remark 5.1 (Single-state notation convention). Within the restricted notation of this paper, an effective vacuum transition is represented, when possible, as a phase or effective-sector rewrite relative to the fixed source datum $(\mathcal{A}_U, \Omega)$, not as $\Omega \to \Omega'$. This is a convention fixing the theorem's inputs, not a proposition that all physical vacuum sectors share one representation.

In AQFT, distinct vacua or phases may give unitarily inequivalent representations. Such a transition may require changing representation/sector data and then lies outside the single-source convention used by this paper; it cannot be ruled out by notation. In ordinary QFT, false-vacuum decay is described through tunnelling between field configurations. Where a single effective representation suffices, the programme-level translation is:

> Vacuum decay may be represented as an effective-sector phase rewrite relative to the fixed source datum, in which the mass spectrum and interaction scales of the read-out may cease to be maintained in their prior form.

The expanding bubble wall in such a transition is normally described as approaching light-like propagation in the effective spacetime. That statement belongs to the effective Lorentzian geometry. It does not imply that $\mathcal{A}_U$ changes, nor that Ω splits.

6 Layer Separation

This paper deliberately keeps its theorem layer and its programme-level reading layer separate. Section 5 is not part of the theorem chain; it is an interpretation layer built on the theorem after the standard Lorentzian result has been recovered.

Layer	**Claim**	**Status**
Theorem layer	Given effective Lorentzian geometry and admissible P^μ, the mass–energy relation follows	Conditional theorem
Bridge layer	$\hat{P}_\mu$ is the Stone generator of admissible local translation unitaries on $\mathcal{R}_{\text{adm}}$; P^μ is its c-number read-out	Bridge assumption plus consistency checks
Structural reading layer	c^2, mass, energy, Higgs/QCD contributions, and vacuum sectors can be read as read-out or imprint features	Programme-level interpretation

The separation is necessary. If all claims were presented as theorem-level, the paper would overstate itself. If all claims were presented as interpretation, the real theorem-level content would be undersold. The correct position is mixed: a narrow theorem supported by an explicitly non-theoremic structural interpretation.

6.1 Mass as timelike stabilization

The following statement is interpretive:

> Positive invariant mass may be read as a measure of how an admissible stable structure is constrained to carry a timelike momentum norm inside the effective Lorentzian read-out.

This reading is compatible with Theorem 4.3, but it is not a theorem beyond it. It does not say that mass is a substance. It does not say that massive systems are metaphysically inferior to massless ones. It says only that, in a Lorentzian sector, timelike and null read-outs differ in their invariant norm structure.

6.2 Massless propagation as null read-out

Corollary 4.4 gives the theorem-level fact: if $m = 0$, then P^μ is null and $E = |\mathbf{p}|c$. A programme-level reading may say that a massless excitation lies on the boundary where temporal and spatial contributions to the Lorentzian norm cancel. That reading must remain a reading. It does not identify photons with any deeper source object. It does not turn light into a metaphysical first principle inside the paper.

Nor does Lorentzian geometry by itself prove that a particular massless particle species must exist in the effective field theory. The theorem supplies the null kinematic class; whether a physical excitation occupies that class is a dynamical and field-theoretic question. In the Standard Model, the photon's masslessness is tied to gauge structure and electroweak symmetry breaking, not to Lorentzian signature alone.

6.3 Read-out without identity

The general guardrail is:

> A structure may carry or display an imprint of a deeper constraint without being identical to that constraint.

Thus c^2 can carry the metric's temporal-spatial scale in the mass–energy relation without being the metric itself. Higgs/Yukawa and QCD mechanisms can contribute to the same invariant-mass bookkeeping without being the same mechanism. A null four-momentum can mark the light-cone structure without being the source algebra. This is the difference between read-out and identity.

7 Discussion

7.1 Why this is a conditional convention check

Theorem 4.3 records a conditional Lorentzian identity. Once an effective Lorentzian metric, translation/Stone bridge, four-vector P^μ, and invariant-mass definition are supplied, the standard mass–energy relation follows by convention-level algebra. The theorem's role is to record sufficiency and fix signs and factors of c; the physical content is entirely in the bridge.

This framing avoids claiming a derivation of special relativity or mass-generating physics. It also makes the limited architectural point explicit: the identity enters only after the programme assumes the same Lorentzian and momentum structures used in the standard derivation.

7.2 Why the bridge definition matters

The definition of admissible four-momentum is the load-bearing step. Without it, one could only say that a Lorentzian read-out has been assumed or modelled. With it, one can state relativistic energy and mass relations inside the read-out. Because the definition is explicit, reviewers can evaluate it directly:

- Is an admissible local translation action available in the sector under discussion?
- Is it implemented by a strongly continuous unitary representation on the effective read-out Hilbert space?
- Are the Stone generators well-defined on a domain adequate for the local calculation?
- Does the object reduce to standard four-momentum in the appropriate limit?

Those are the right questions. They are more precise than asking whether modular flow alone somehow "is" momentum.

7.3 Relation to standard relativity, QFT, and AQFT

The theorem-level calculation in this paper is standard special-relativistic kinematics in a local Lorentzian frame. In that respect it is continuous with textbook treatments of mass–energy equivalence and relativistic four-momentum [1, 2, 3]. It also leaves untouched the Standard Model mechanisms that generate elementary and composite mass scales, including Higgs/Yukawa mass generation and QCD contributions to nucleon mass [12, 13, 15, 16].

The algebraic background is closer in spirit to algebraic QFT, where observables are represented by operator algebras and spacetime symmetries, when present, are implemented by automorphism groups and unitary representations [4]. Tomita–Takesaki modular theory and related results such as the Bisognano–Wichmann theorem show that modular data can carry geometric significance in relativistic quantum theory [5, 8, 9]. The present paper does not identify modular flow with physical momentum. Instead, it assumes a Lorentzian read-out and separately requires local translation unitaries before defining momentum via Stone generators.

This is also the point at which Stone's theorem enters: strong continuity of the local translation unitaries is the technical condition that permits self-adjoint momentum generators [6, 7]. In curved or locally covariant settings, translation symmetry is generally local, approximate, or replaced by more general covariance data. Locally covariant QFT makes this dependence on background/category data explicit, while dynamical locality sharpens when the theory responds consistently to spacetime changes [10, 11]. Neither framework automatically supplies the translation representation assumed here.

Thus the paper's contribution is not a new QFT computation or an alternative to AQFT. It is a conditional placement statement: if a Lorentzian read-out and the usual local translation representation are supplied, then the standard mass–energy identity holds inside that read-out.

7.4 Future strengthening

A future paper could try to upgrade the bridge definition into a derived lemma by deriving the Lorentzian read-out rather than supplying it, and then showing that a canonical momentum structure follows on admissible translation sectors. That would be a stronger result, but it would be a different paper. It would require a direct analysis of translation symmetries, generators, and conjugate variables inside the algebraic read-out construction.

The present paper does not need that stronger claim. Its narrower theorem is already useful because it closes a specific architectural gap: it shows how mass–energy equivalence sits in the programme without smuggling in an unmarked physical object.

8 Limits

The following limits should be read as part of the result, not as afterthoughts.

1. **No new numerical prediction.** The paper does not predict particle masses, coupling constants, or the value of c.
2. **No replacement of SR/QFT.** The theorem uses standard Lorentzian geometry and is compatible with standard relativistic physics.
3. **No derivation of the Higgs mechanism.** Higgs/Yukawa mass generation is used as physical anchoring, not derived from Q-RAIF A.
4. **No derivation of QCD confinement.** QCD contributions to nucleon mass are acknowledged as standard physics, not replaced by read-out language.
5. **No multi-Ω drift.** Effective phase or vacuum-sector changes are not represented as changes of the global state Ω.
6. **No identity collapse.** Read-out, imprint, and structural interpretation do not license statements of identity between an emergent trace and its generating condition.
7. **Local/curved limitation.** The identity is stated in local orthonormal frames or local/asymptotic regimes admitting an effective translation representation. In a generic curved effective geometry, global conserved energy requires additional structure such as a Killing symmetry or an asymptotic time translation.
8. **Open Type III bridge.** Appendix B verifies the translation/Stone conventions only in flat Minkowski space. Existence of a suitable representation and common generator domain in an emergent Type III_1 read-out sector is not established.

These limits preserve the theorem's usable strength. The paper is strongest when it is read as a controlled compatibility result with a disciplined interpretive layer.

A Detailed Convention Check

Two equivalent conventions are common.

A.1 Coordinate-time convention

Use coordinates (t, x, y, z) and metric

$$ds^2 = -c^2dt^2 + dx^2 + dy^2 + dz^2.$$

In this convention the components of the metric carry explicit factors of c^2.

A.2 Length-time convention

Use $x^0 = ct$ and write

$$ds^2 = -(dx^0)^2 + dx^2 + dy^2 + dz^2.$$

Then

$$g_{\mu\nu} = \mathrm{diag}(-1, 1, 1, 1), \qquad P^\mu = \left(\frac{E}{c}, \mathbf{p}\right).$$

This is the convention used in the proof of Theorem 4.3. It makes the invariant relation

$$m^2c^2 = -g(P, P)$$

dimensionally transparent.

A.3 Observer normalization

If u^μ is dimensionless and satisfies $g(u,u) = -1$, then

$$E_u = -c\, g(P,u).$$

If instead U^μ is a physical four-velocity satisfying $g(U,U) = -c^2$, then

$$E_U = -g(P,U).$$

The two formulas are equivalent under $U = cu$. The main text uses the dimensionless convention.

B Standard-Limit Recovery

The admissible four-momentum definition must recover ordinary special relativity in the appropriate local or large-system limit. In flat spacetime, translation invariance gives generators

$$\partial_\mu = \frac{\partial}{\partial x^\mu}.$$

The corresponding strongly continuous translation unitaries are

$$U(a) = \exp\left(-\frac{i}{\hbar} a^\mu \hat{P}_\mu\right),$$

so the momentum operators are their self-adjoint Stone generators. In the usual coordinate representation,

$$\hat{P}_\mu = -i\hbar\, \partial_\mu,$$

with eigenvalues P_μ for momentum eigenstates. Classically, the corresponding four-momentum is

$$P^\mu = m\frac{dx^\mu}{d\tau}$$

for a massive particle, and a null momentum vector for massless excitations. In both cases the invariant relation is the mass shell condition

$$g(P,P) = -m^2c^2.$$

Definition 3.3 is designed to reduce to this standard structure when the admissible read-out sector becomes the usual local relativistic sector. This appendix verifies only that standard-limit compatibility. It does not establish the existence, common-domain, or covariance conditions of Assumption 3.4 in a genuinely emergent or Type III_1 sector.

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